Syed Hussam Abbas

Plant Molecular Biologist | RNA Surveillance (NMD), Heat Stress Tolerance and Thermomemory
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RESEARCH INTERESTS

Post-transcriptional regulation of gene expression in plants, with a focus on nonsense-mediated mRNA decay (NMD), alternative splicing, heat shock factor (HSF) biology, and thermomemory in *Arabidopsis thaliana*; building on earlier work on transcription fidelity and alternative splicing control. My broader interest is in the productive transcriptome, the fraction of transcripts that escape surveillance and are actually translated, and in how post-transcriptional and translational quality control jointly set it during stress exposure and recovery. A current direction is to treat RNA surveillance capacity as a quantitative trait that may vary between Arabidopsis accessions and shape thermal acclimation. I am also interested in how plants sense environmental signals through phytochromes, cryptochromes and hormone signalling, and in the proteomics of the protein complexes.

EDUCATION & RESEARCH

PhD in Plant Sciences Oct 2022 – Mar 2027

Hungarian University of Agriculture and Life Sciences (MATE), Gödöllő, Hungary

Supervisor: Dr. Tibor Csorba, Plant Stress Biology Group

Thesis: "Transcriptional and post-transcriptional regulation of heat stress response and memory in plants"

- Showed that nonsense-mediated mRNA decay (NMD) ambivalently regulates heat stress memory in *Arabidopsis*, balancing HSFA2 splice isoforms during acclimation and promoting recovery *(*Abbas et al., *bioRxiv*, 2026*)*
- Investigated how transcription elongation factors TFIIS and IWS1 coordinate transcription and co-transcriptional RNA processing under heat stress *(manuscript in preparation)*
- Contributed RT-qPCR, genotyping, western blotting and mycophenolic acid (MPA) assays to defining a cooperative TFIIS/NMD transcriptome quality control pathway (Szaker et al., *The Plant Cell*, 2025)
- Contributed western blotting and FPLC protein complex purification to a study on DNA methylation and trichome development in tomato (Nyikó et al., *The Plant Journal*, 2025)
- Optimized the long-term acquired thermotolerance (LAT) phenotyping regime and memory-scoring assays now used routinely in the group
- Dissected temperature-dependent splicing and NMD-sensitive transcript dynamics in *upf* mutants using isoform-specific RT-qPCR
- Characterized stress phenotypes using confocal and fluorescence microscopy, SDS-PAGE and western blotting

MSc in Plant Breeding and Genetics Sep 2019 – Jul 2022

University of Agriculture Faisalabad, Pakistan

Thesis: "Assessment of association among different yield contributing factors in maize (*Zea mays* L.) germplasm"

- Quantitative genetic analysis of trait associations across a maize germplasm panel, including correlation and path coefficient analysis of yield components
- Supported molecular biology experiments including nucleic acid extraction, PCR-based genotyping, and sample preparation

PUBLICATIONS

1. **Abbas S. H.**, Szaker H. M., Darkó É., Erős T., Dániel Silhavy and Csorba T. (2026) Ambivalent roles of nonsense-mediated mRNA decay in heat stress tolerance. *bioRxiv*. https://doi.org/10.64898/2026.10.30.763111
2. Szaker H. M.*, Verma R.*, Szádeczky-Kardoss I., Gál N., **Abbas S. H.**, Darkó É., Pettkó-Szandtner A., Silhavy D. and Csorba T. (2025) Transcription fidelity and control of alternative splicing contribute to heat stress survival in Arabidopsis. ***The Plant Cell***, *37(11): koaf256.* https://doi.org/10.1093/plcell/koaf256
3. Nyikó T., Gyula P., Ráth S., Sós-Hegedűs A., Csorba T., **Abbas S. H.**, Bóka K., Pettkó-Szandtner A., Móricz Á. M., Molnár B. P., Erdei A. L. and Szittya G. (2025) INCREASED DNA METHYLATION 3 forms a potential chromatin remodelling complex with HAIRPLUS to regulate DNA methylation and trichome development in tomato. ***The Plant Journal***, *121: e70085*. https://doi.org/10.1111/tpj.70085
4. Singh K., **Abbas S. H.**, Singh S., Iqbal N., Kocsy G., Pérez-Pérez J. M., Riyazuddin R. (2025) Deciphering the Plants' Response to Flood-Induced Hypoxia and Anoxia at Molecular Level. In: Riyazuddin R., Gupta R., Ramteke P. W., Sayyed R. (eds) Plant Flooding. Plant in Challenging Environments. Springer, Cham, 6:93-112. https://doi.org/10.1007/978-3-031-83068-6_5

Full list of publications can be found here: https://syedhussamabbas.github.io/#publications

TECHNICAL SKILLS

RNA & Molecular Biology
- RT-qPCR and isoform-specific qPCR, PCR
- Molecular cloning, DNA/RNA extraction and plasmid preparation
- Design of heat stress assays for short- and long-acquired thermotolerance (SAT, LAT, TMHT), acclimation, recovery and thermomemory

Nuclear Biology & Chromatin
- Nuclei isolation from sample tissues
- Nuclear protein extraction and sample preparation for proteomics
- Histone extraction for post-translational modification analysis

Protein Biochemistry
- Western blotting, SDS-PAGE, co-immunoprecipitation (Co-IP)
- Recombinant protein purification by FPLC (affinity/size-exclusion)
- Biochemical characterization of protein complexes
- Subcellular fractionation (nuclear and cytoplasmic)

Microscopy & Imaging
- Fluorescence microscopy (Leica) and confocal laser scanning microscopy
- Subcellular localization of fluorescent protein fusions
- Optical (bright-field) microscopy

Bioinformatics & Data Analysis
- RNA-seq data interpretation, differential expression and splicing analysis
- R programming for statistical analysis and data visualization
- Familiarity with plant genome databases (TAIR, Ensembl Plants)

Plant Biology
- *Arabidopsis thaliana* cultivation, transformation, genotyping and phenotyping
- Controlled abiotic stress experiments (heat, hypoxia/anoxia) in growth chambers

ADDITIONAL EXPERIENCE

Campus Mentor Jul 2024 – Jun 2026

Stipendium Hungaricum Mentor Network (HOOK), Hungary

- Mentored international PhD and MSc students on academic integration, administrative procedures and personal well-being, easing their transition to a new academic and cultural environment
- Guided students through university procedures, documentation and local settling, and served as primary point of contact with university services
- Organized student activities and social events, monitored academic progress and well-being, and prepared structured monthly reports for program coordination

HONORS & AWARDS

- Best Poster Award (de.NBI 2026: Bioinformatics for Proteomics, RUB Bochum, Germany)
- SH Dissertation Scholarship (Hungarian government scholarship for international doctoral researchers)
- Best Publication Award (GBI Day 2025, UniMATE, Hungary)
- Best Poster Award (E^3UDRES^2 Science Festival 2025, Latvia)
- Stipendium Hungaricum Scholarship (Hungarian government scholarship for doctoral researchers)
- Pannónia Mobility Scholarship (Hungarian mobility award for international research activities)
- Chief Minister's Merit Scholarship (competitive academic merit award, Pakistan)

REFERENCES

References available upon request.